Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

Project Workflow: Week 3

Project: EasyStego: A Drag-and-Drop Image Steganography Tool
Phase: Environment Setup & Project Initialization
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Course: Software Development I (CSE 2216)
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1. Objective

The primary goal of Week 3 was to prepare a fully functional development environment and initialize the EasyStego project structure. This included:

- Installing and verifying Java JDK 11+ as the core runtime and compiler.
- Configuring Visual Studio Code as the IDE with the Java Extension Pack.
- Setting up Git and creating the GitHub repository for version control.
- Creating the initial project file structure with the main EasyStego class extending JFrame.
- Verifying that a basic Java Swing window launches correctly before development begins.

2. Detailed Task Breakdown

JDK Installation and Verification: Downloaded and installed Java Development Kit (JDK) 11+ from the official Oracle or OpenJDK distribution. Verified the installation by running java -version and javac -version in the terminal to confirm the correct version is active and the compiler is on the system PATH.

IDE Configuration: Installed Visual Studio Code and set up the Java Extension Pack (including Language Support for Java, Debugger for Java, and Test Runner for Java). Configured the workspace settings to point to the correct JDK installation. Verified IntelliSense, error highlighting, and the integrated terminal are working correctly.

Git and GitHub Setup: Initialized a local Git repository in the project folder using git init. Created a new public GitHub repository named EasyStego. Linked the local repo to the remote with git remote add origin. Pushed the initial commit containing the project structure and README.md.

Project Structure Creation: Created the main source file EasyStego.java in the project root. Defined the EasyStego class extending JFrame as the top-level application container. Added the main() method calling SwingUtilities.invokeLater() to ensure the GUI runs on the Event Dispatch Thread (EDT) as required by Java Swing.

Initial Swing Window Verification: Compiled EasyStego.java using javac and ran the resulting class file with java EasyStego. Confirmed that the application window launches correctly with the correct title, size (900×650), and is positioned at the center of the screen. This verified the environment is fully ready for GUI development in Week 4.

3. Project Structure Initialized

The following project structure was created during Week 3:

File / Folder	Purpose
EasyStego.java	Main application class — extends JFrame, entry point
README.md	Project documentation (completed in Week 2)
.gitignore	Excludes compiled .class files and IDE config folders
secret_image.png (sample)	Placeholder test image for development and testing

4. Deliverables

- Functional development environment: JDK 11+, VS Code, and Java Extension Pack installed and verified.
- GitHub repository created and linked to local project folder.
- Initial project structure committed and pushed to GitHub.
- EasyStego.java: main class with JFrame setup, setTitle(), setSize(), setLocationRelativeTo(), and main() method verified and launching correctly.
- .gitignore file configured to exclude compiled output.

5. Tools & Resources

Development Environment:

- Java Development Kit (JDK) 11+ — Oracle or OpenJDK distribution
- Visual Studio Code — IDE for Java development
- Java Extension Pack for VS Code (Language Support, Debugger, Test Runner)
- Git — local version control
- GitHub — remote repository hosting

Terminal Commands Used:

- java -version and javac -version — verify JDK installation
- git init — initialize local repository
- git remote add origin <url> — link to GitHub remote
- git add . && git commit -m "Initial commit" — first commit
- git push -u origin main — push to GitHub
- javac EasyStego.java — compile source file
- java EasyStego — run and verify the application window